MIL-B-46413C (MU)
29 JULY 1966
SUPERSEDING
MIL-B-46413B (MU)
25 JANUARY 1963

MILITARY SPECIFICATION

BURSTER, PROJECTILE, M71 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing of one burster designated as Burster, Projectile M71.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for

MIL-R-11470 — Radiographic Inspection; Qualification of Equipment, Operators and Procedures

MIL-R-11471 — Radiographic Inspection of Metals

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

MIL-STD-109 — Quality Assurance Terms and Definitions

MIL-STD-1168 — Lot Numbering of Ammunition

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

8861032 — Burster, Projectile, M71

8861837 — Box, Packing, Ammunition for Burster Projectile M71, with Loading Accessories

FSC 1320

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8861838 — Container, Ammunition Fiber, for Burster Projectile, M71, with Loading Accessories

PUBLICATIONS

ARMY

ET–8861032 — Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Burster. The burster shall comply with all requirements specified on Drawing (dwg.) 8861032, and with all requirements specified in applicable specifications.

3.3 Cavitation.

3.3.1 *Area from the bottom to one inch above bottom.* There shall be no single cavity having a projected length of more than 1/8 inch or a sum total of cavities of more than 1/64 square inch. There shall be no cracks or annular rings in the charge in this area.

3.3.2 *Area from top to one inch from bottom of container.* There shall be no single cavity having a projected length of more than one inch or a sum total of cavities of more than 3/8 square inch and there shall be no more than three transverse cracks 1/32 inch wide, maximum, in this area.

3.3.3 *Separation of charge from base of container.* There shall be no void greater than 1/32 inch between the charge and the base of the container.

3.3.4 *Porosity.* Porous areas shall be treated as cavities, and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purpose.

3.4 Workmanship. All parts shall be free from chips, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

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(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL–STD–105 and shall be numbered in accordance with MIL–STD–1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor. Each lot of bursters shall contain:

(a) Composition B4 from not more than one lot.

(b) Parts from one manufacturer from not more than one lot interfix number.

(c) Accessories (supplementary charge assembly and projectile burster support) from not more than one lot interfix number.

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4.2.2 *Examination.* Inspection for Critical defects (and Major defects, when so specified,) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with Standard MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Container, burster, prior to painting (see dwg. 8861033 covering a detail of dwg. 8861032).

Categories	*Defects*	*Method of Inspection*	*Code No. (see 6.2)*
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 4.0 percent		
201.	Total length, minimum (min.)	Gage	01001
202.	Thickness through bottom, min	Gage	01002
203.	Diameter, min	Gage	01003
204.	Wall thickness	Gage	01004
205.	Radius of spherical indents	Gage	01005
206.	Depth of indents, min	Gage	01006
207.	Insufficient number or incorrect location of indents	Visual-Scale	01007
208.	Evidence of poor workmanship (see 3.4)	Visual	01008

4.2.2.2 Container, burster (see dwg. 8861033 covering a detail of dwg. 8861032).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Interior surface not completely painted	Visual	02001
102.	Foreign matter in interior	Visual	02002
103.	Pool of paint	Visual	02003
Minor:	None defined.		

4.2.2.3 Assembly (see dwg. 8861032).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:			
1.	Cushioning pad missing or insecure	Visual	03001

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Categories	*Defects*	*Method of Inspection*	*Code No.*
2.	Closure missing	Visual	03002
3.	Burster container cracked	Visual	03003
4.	Explosive charge loose in container (see Note 1. below)	Manual	03004

Note 1. The inspection for loose charge shall be conducted by pressing against the charge with the thumb and by inverting the tube and tapping it against the hand. The assembly shall be classed as defective if there is any perceptible movement of the charge during this test.

Major:	AQL 0.65 percent		
101.	Failure to chamber gage	Gage	03005
102.	Total length, maximum (without compression of cushioning pad)	Gage	03006
103.	"Up" arrow points in wrong direction	Visual	03007
Minor:	AQL 1.50 percent		
201.	Closing disc ruptured or incompletely sealed	Visual	03008
202.	Evidence of poor workmanship (see 3.4)	Visual	03009
203.	Cushioning pad not centered	Visual	03010
204.	Marking misleading or unidentifiable, including incorrect color code	Visual	03011

4.2.2.4 Container prior to sealing (see dwg. 8861838).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Burster projectile missing	Visual	04001
Minor:	AQL 0.65 percent		
201.	O-ring missing	Visual	04002

4.2.2.5 Container sealed (see dwg. 8861838).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Tape improperly applied	Visual	05001
Minor:	AQL 1.00 percent		
201.	Marking misleading or unidentifiable	Visual	05002
202.	Contents loose upon shaking	Manual	05003

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4.2.2.6 Sealed wood packing box (see dwg. 8861837).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Box damaged	Visual	06001
102.	Strapping missing, broken or loose	Visual-Manual	06002
Minor:	AQL 1.50 percent		
201.	Strapping improperly assembled	Visual	06003
202.	Lid improperly assembled	Visual	06004
203.	Markings misleading or unidentifiable	Visual	06005
204.	Contents loose in box	Manual	06006

4.2.3 *Testing.*

4.2.3.1 *Cavitation (see 3.3).* Cavitation defects are classified as follows:

Critical — (Code No. 03012)

Any cavity, crack or annular ring in the area from the bottom to one inch above the bottom (see 3.3.1)

Any separation of charge from the base of the container in excess of 1/32 inch (see 3.3.3).

Major — (Code No. 03013)

Cavitation or cracks in excess of that permitted by 3.3.2 and 3.3.4.

The sampling plan and procedure for this test for both critical and major defects shall be in accordance with Standard MIL-STD-1235 using a CSP-2 plan having an AQL of 0.25 percent except that when a critical defect is found, 100 percent inspection for i successive defect free units (for both criticals and majors) shall be required immediately instead of continuing on a sample basis. When a major defect is found during the f phase of the plan, inspection shall continue at the same frequency until the homogeneity requirement is fulfilled or 100 percent inspection is required. Appropriate i and f values shall be as specified by the procuring activity. The test shall be performed as specified in 4.3.1.

4.2.4 *Inspection Equipment.* Equipment Tabulation Number ET-8861032 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

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4.2.4.3 *Government use of contratctor's inspection equipment.* The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Determination of cavities.* The sample bursters shall be subjected to x-ray examination. In all exposures the x-ray beam shall form a 90 degree angle with the longitudinal axis of the projectile. Examination of the x-ray negative shall be made to determine compliance with the applicable requirements.

Note. Radiographic equipment, operations and procedures shall be qualified in accordance with Specification MIL-R-11470. In addition, the sensitivity of x-ray technique shall be 2.0 percent, and the photographic density of the film shall be no less than 1.25 as determined in accordance with 4.3.1.1 and 4.3.1.2. In the event an original x-ray picture leaves doubt as to acceptability, the burster shall be re-x-rayed at an angle 90 degrees from that of the original exposure and in the same plane.

4.3.1.1 *Check of sensitivity of x-ray technique.* The sensitivity of x-ray technique shall be determined by means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the burster away from the film (source side). Observation shall be made of the image of the penetrameter on the x-ray film. The sensitivity of x-ray technique shall be considered satisfactory when the image of the penetrameter, including the 2t hole, is sharply defined on the x-ray film. Unless otherwise specified, the penetrameter shall be made of steel having the approximate specific gravity of 7.8. The thickness of the penetrameter shall be based on a combination of both charge and steel thickness and shall be equal to 2.0 percent of the maximum equivalent thickness. The penetrameter shall have a hole, the diameter of which is twice the thickness of the penetrameter. For further details on construction of the penetrameter see Specification MIL-R-11471. The actual thicknesses may be determined from a radiograph of the metal parts. The penetrameter thickness shall be determined as illustrated in Figure 1 of this specification.

4.3.1.2 *Check of photographic density of film.* The photographic density of the film shall be determined on the film in a region of sound charge plus metal corresponding to that portion of the material which is of the greatest combined thickness to be examined on the negative. To judge densities, it is recommended that standards of radiographic densities be used (see 6.4). Comparison of film with standards may be made visually; however, the use of a densitometer is recommended.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* The M71 projectile burster shall be preserved and packaged in accordance with dwg. 8861838.

5.2 Packing.

5.2.1 *Level A.* The cartons shall be packed in accordance with dwg. 8861837.

5.3 Marking. The container shall be marked in accordance with dwg. 88861838. The wood box shall be marked in accordance with dwg. 8861837.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) *Data cards.* Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

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6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (SMUPA Form 1010) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the EL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL–D–45608.

CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

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6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35mm microfilm process. Contractors shall submit one copy of the final design as a rolled set or flat set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 *Submission of contractor designs.* Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Film density. "American Standard for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, N. Y., has been found satisfactory as a source of information on film densities.

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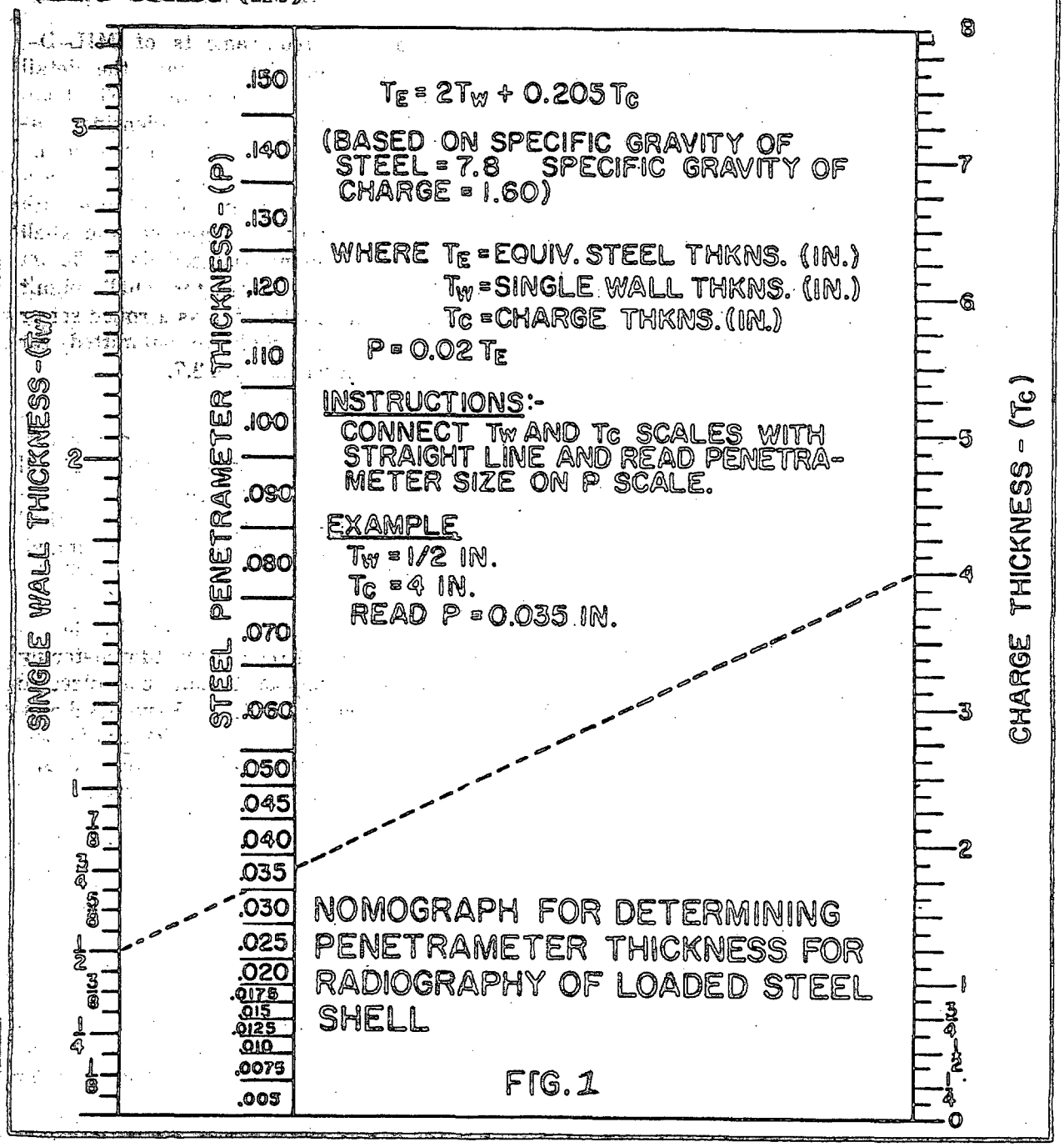
Army—MU

Preparing activity:

Army—MU

Project No. 1320-A111

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NOMOGRAPH FOR DETERMINING PENETRAMETER THICKNESS FOR RADIOGRAPHY OF LOADED STEEL SHELL

FIG. 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

DIRECT GOVERNMENT CONTRACT SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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